OCEAN EXPLORER

Field Journal

Dive deep. Look close. Predict the zone.



SPECIMEN LABEL

Ocean scientist: _____

Journal started: _____ **Animals logged:** _____

If found, please return to the DDMES Science Olympiad binder shelf.

How an ocean explorer works

Scientists who study the ocean are called **oceanographers** and **marine biologists**. Most of the deep sea has never been seen by people, so every picture is a clue! This journal lives in your SciOly binder.

1
FIND
Aquariums, museums, the beach, books, and ocean videos all count.

2
LOOK
Eyes, color, mouth, body, lights, fins, and shell.

3
PREDICT
Guess its zone from its body clues before you look it up.

4
CHECK
Find the real answer. Were you right? Why?

5
LOG IT
Fill in an Ocean Animal page and your Life List.

The Field Scientist Code

- [ ] At the beach, look but don't take living animals. Put rocks back.
- [ ] Never touch jellyfish, even ones washed up on the sand.
- [ ] Swim and wade only with an adult watching.
- [ ] Watch ocean videos with an adult (deep-sea cameras from NOAA and MBARI are great).
- [ ] Leave shells with animals living inside them.
- [ ] Wash your hands after tide pooling.

My field kit

- [ ] This journal and a pencil
- [ ] Colored pencils (blues and a dark color)
- [ ] Hand lens
- [ ] Your zone and animal flashcard ring
- [ ] A bucket for looking, not keeping
- [ ] A tablet for ocean videos (with an adult)

Journal map

Tools and guides	p. 3-6
Ocean Animal pages	p. 7-12
Hunts and missions	p. 13-19
Life List and credits	p. 20-21

Where to look

The museum (Nov 7)
living and preserved ocean animals, whale skeletons, and ocean dioramas

NC aquariums
Fort Fisher, Pine Knoll Shores, and Roanoke Island

The beach
shells, crabs, barnacles, jellyfish, sea stars, seaweed

Deep-sea videos
NOAA Ocean Exploration and MBARI livestreams of real dives

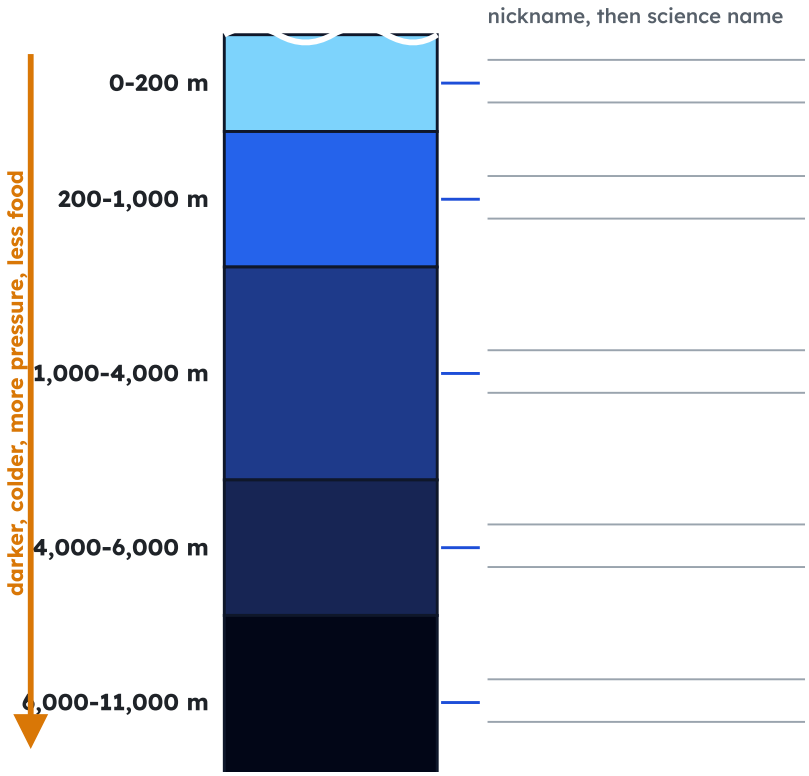
Books
library books about the deep sea and ocean zones

A fish store
clownfish, crabs, and other ocean animals

Did you know? The ocean's deepest spot, the Challenger Deep, is about 11,000 meters down. If you put Mount Everest there, its top would still be more than 2,000 meters under water!

The ocean column

The ocean has layers, like floors in a very tall building. Write each zone's nickname and science name. The depths are there to help you.



Sunlight **epipelagic** **Twilight** **mesopelagic** **Midnight** **bathypelagic** **Abyss** **abyssopelagic** **Trenches** **hadalpelagic**

Compare the zones

	Top	Bottom
Sunlight		
Temperature		
Pressure		
Food		

Write more or less, warm or cold, low or high.

Across the ocean

Intertidal: the beach zone where tides cover and uncover the land. **Neritic**: shallow water over the continental shelf (most fishing happens here). **Oceanic**: the open ocean past the shelf, where the deep zones are.

Ocean Animal Collector

Every animal from our lessons. Check the box when you see one at an aquarium, the museum, the beach, or in a real video.



Clownfish

Lives in a sea anemone for protection.

Sunlight ☐ Spotted: _____



Sea turtle

Flippers for long swims and countershading.

Sunlight ☐ Spotted: _____



Tuna

Fast torpedo shape and countershading.

Sunlight ☐ Spotted: _____



Dolphin

Blowhole to breathe air.

Sunlight ☐ Spotted: _____



Jellyfish

See-through body.

Sunlight ☐ Spotted: _____



Zooplankton

Tiny drifter; food for almost everything.

Sunlight ☐ Spotted: _____



Lanternfish

Belly lights (counter-illumination).

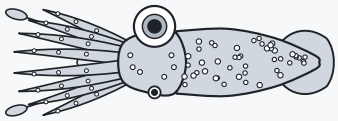
Twilight ☐ Spotted: _____



Hatchetfish

Belly lights and silver sides.

Twilight ☐ Spotted: _____



Squid

Big eyes and can glow.

Twilight ☐ Spotted: _____



Swordfish

Heats its eyes and brain to see and think in cold water.

Twilight ☐ Spotted:



Octopus

Changes color and squeezes into small spaces.

Twilight ☐ Spotted:



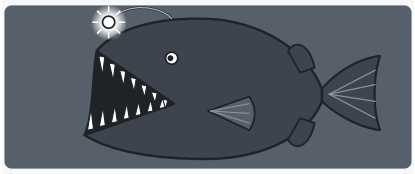
Deep-sea red crab

Red color looks black in deep water.

Twilight ☐ Spotted:

Ocean Animal Collector, continued

The deep animals and the intertidal zone.



Anglerfish

Glowing lure.

Midnight ☐ Spotted: _____



Gulper eel

Huge mouth and stretchy stomach.

Midnight ☐ Spotted: _____



Viperfish

Long fangs and a light lure.

Midnight ☐ Spotted: _____



Sea cucumber

Eats marine snow from the sea floor mud.

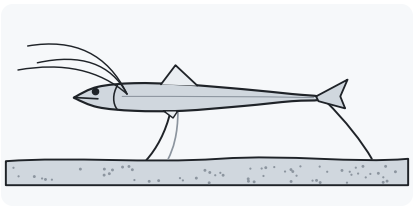
Abyss ☐ Spotted: _____



Brittle star

Long bendy arms to grab food.

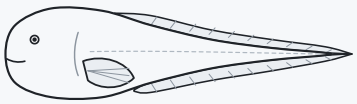
Abyss ☐ Spotted: _____



Tripod fish

Stands on 3 long fins and waits for food drifting by.

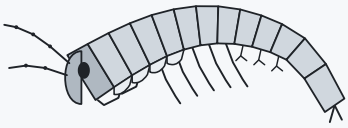
Abyss ☐ Spotted: _____



Snailfish

Soft jelly-like body handles pressure.

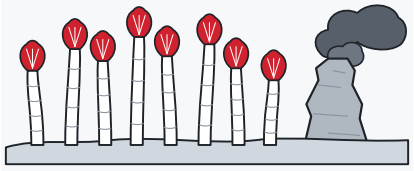
Trenches ☐ Spotted: _____



Amphipod

Scavenges dead animals that sink.

Trenches ☐ Spotted: _____



Tube worm

Bacteria inside make its food.

Trenches ☐ Spotted: _____



Barnacle

Closes its shell plates at low tide to keep water in.

Intertidal ☐ Spotted: _____

Zone colors

Sunlight 0-200 m

Twilight 200-1,000 m

Midnight 1,000-4,000 m

Abyss 4,000-6,000 m

Trenches 6,000-11,000 m

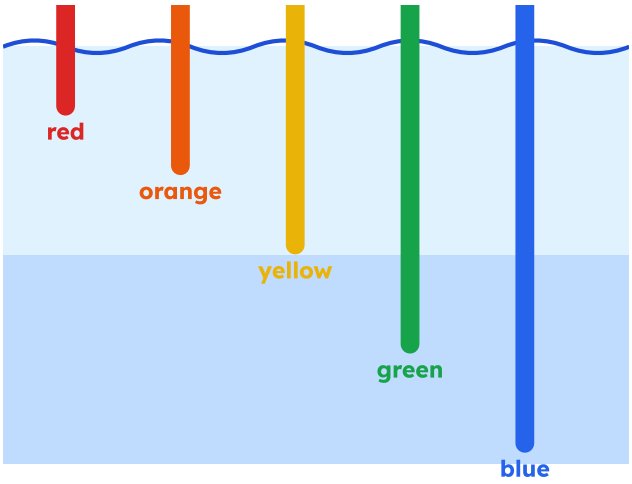
Intertidal where tides go in and out

Adaptation Detective

An **adaptation** is a body part or behavior that helps an animal survive. Draw a line from each body clue to how it helps.

Huge eyes	●	● handles crushing pressure
Red or black body	●	● is hard to see in open water
Countershading (dark back, light belly)	●	● can gulp any food it finds, since meals are rare
Lights on its belly	●	● catches the tiny bit of light in dim water
Huge mouth and stretchy stomach	●	● hides its shadow from hunters below
Soft, watery body	●	● blends in when seen from above or below
See-through body	●	● is invisible where red light can't reach

Why red disappears first



Colors fade with depth

Sunlight is made of all colors. Water soaks up **red** first, then orange and yellow. Blue light goes deepest.

So a red animal in the deep sea looks **black**. It is hidden!

Which animal in our collector uses this trick?

_____

_____

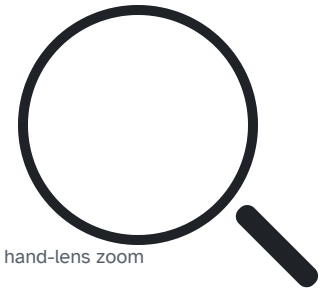
Ocean Animal #1

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

Date _____ **Place or video** _____

Where I saw it: ☐ aquarium ☐ museum ☐ beach ☐ video ☐ book ☐ other

Sketch it BIG. Label 3 adaptations.



hand-lens zoom

Real size: about _____ cm

BODY CLUES

- ☐ big eyes
- ☐ tiny or no eyes
- ☐ red or black
- ☐ see-through
- ☐ silver sides
- ☐ countershaded
- ☐ glows
- ☐ huge mouth
- ☐ soft body
- ☐ shell or armor

IT MOVES BY

- ☐ swimming
- ☐ drifting
- ☐ crawling
- ☐ staying put

MY ZONE GUESS

- ☐ sunlight
- ☐ twilight
- ☐ midnight
- ☐ abyss
- ☐ trenches
- ☐ intertidal

ANIMAL _____

REAL ZONE _____

WAS MY GUESS RIGHT? _____

HOW IT GETS FOOD _____

BEST ADAPTATION _____

What was it doing? ☐ eating ☐ hunting ☐ hiding ☐ glowing ☐ swimming ☐ resting ☐ other

Why did I guess that zone?

_____

_____

_____

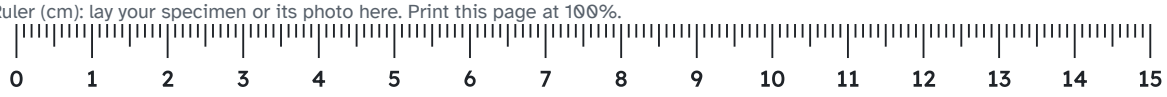
I wonder...

_____

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Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



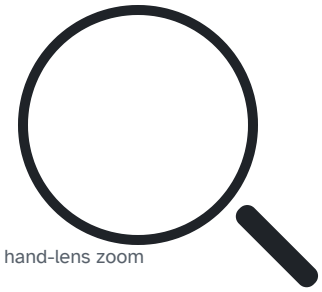
Ocean Animal #2

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

Date _____ **Place or video** _____

Where I saw it: ☐ aquarium ☐ museum ☐ beach ☐ video ☐ book ☐ other

Sketch it BIG. Label 3 adaptations.



hand-lens zoom

Real size: about _____ cm

BODY CLUES

- ☐ big eyes
- ☐ tiny or no eyes
- ☐ red or black
- ☐ see-through
- ☐ silver sides
- ☐ countershaded
- ☐ glows
- ☐ huge mouth
- ☐ soft body
- ☐ shell or armor

IT MOVES BY

- ☐ swimming
- ☐ drifting
- ☐ crawling
- ☐ staying put

MY ZONE GUESS

- ☐ sunlight
- ☐ twilight
- ☐ midnight
- ☐ abyss
- ☐ trenches
- ☐ intertidal

ANIMAL _____

REAL ZONE _____

WAS MY GUESS RIGHT? _____

HOW IT GETS FOOD _____ **BEST ADAPTATION** _____

What was it doing? ☐ eating ☐ hunting ☐ hiding ☐ glowing ☐ swimming ☐ resting ☐ other

Why did I guess that zone?

_____

_____

_____

I wonder...

_____

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Ruler (cm): lay your specimen or its photo here. Print this page at 100%.

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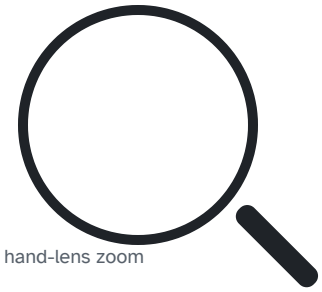
Ocean Animal #3

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

Date _____ **Place or video** _____

Where I saw it: ☐ aquarium ☐ museum ☐ beach ☐ video ☐ book ☐ other

Sketch it BIG. Label 3 adaptations.



hand-lens zoom

Real size: about _____ cm

BODY CLUES

- ☐ big eyes
- ☐ tiny or no eyes
- ☐ red or black
- ☐ see-through
- ☐ silver sides
- ☐ countershaded
- ☐ glows
- ☐ huge mouth
- ☐ soft body
- ☐ shell or armor

IT MOVES BY

- ☐ swimming
- ☐ drifting
- ☐ crawling
- ☐ staying put

MY ZONE GUESS

- ☐ sunlight
- ☐ twilight
- ☐ midnight
- ☐ abyss
- ☐ trenches
- ☐ intertidal

ANIMAL _____

REAL ZONE _____

WAS MY GUESS RIGHT? _____

HOW IT GETS FOOD _____

BEST ADAPTATION _____

What was it doing? ☐ eating ☐ hunting ☐ hiding ☐ glowing ☐ swimming ☐ resting ☐ other

Why did I guess that zone?

_____

_____

_____

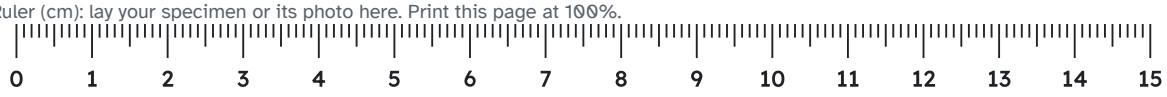
I wonder...

_____

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Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



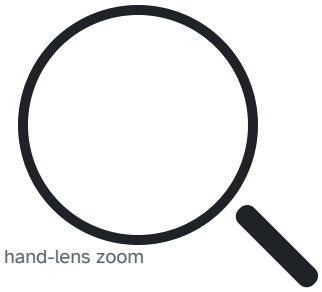
Ocean Animal #4

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

Date _____ **Place or video** _____

Where I saw it: ☐ aquarium ☐ museum ☐ beach ☐ video ☐ book ☐ other

Sketch it BIG. Label 3 adaptations.



hand-lens zoom

Real size: about _____ cm

BODY CLUES

- ☐ big eyes
- ☐ tiny or no eyes
- ☐ red or black
- ☐ see-through
- ☐ silver sides
- ☐ countershaded
- ☐ glows
- ☐ huge mouth
- ☐ soft body
- ☐ shell or armor

IT MOVES BY

- ☐ swimming
- ☐ drifting
- ☐ crawling
- ☐ staying put

MY ZONE GUESS

- ☐ sunlight
- ☐ twilight
- ☐ midnight
- ☐ abyss
- ☐ trenches
- ☐ intertidal

ANIMAL _____

REAL ZONE _____

WAS MY GUESS RIGHT? _____

HOW IT GETS FOOD _____

BEST ADAPTATION _____

What was it doing? ☐ eating ☐ hunting ☐ hiding ☐ glowing ☐ swimming ☐ resting ☐ other

Why did I guess that zone?

_____

_____

_____

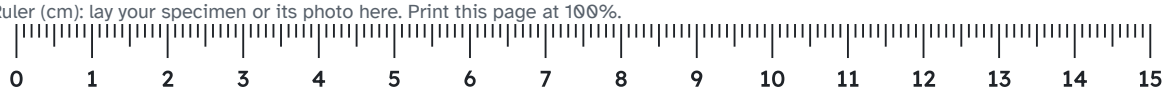
I wonder...

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Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



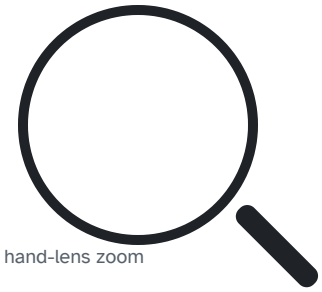
Ocean Animal #5

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

Date _____ **Place or video** _____

Where I saw it: ☐ aquarium ☐ museum ☐ beach ☐ video ☐ book ☐ other

Sketch it BIG. Label 3 adaptations.



hand-lens zoom

Real size: about _____ cm

BODY CLUES

- ☐ big eyes
- ☐ tiny or no eyes
- ☐ red or black
- ☐ see-through
- ☐ silver sides
- ☐ countershaded
- ☐ glows
- ☐ huge mouth
- ☐ soft body
- ☐ shell or armor

IT MOVES BY

- ☐ swimming
- ☐ drifting
- ☐ crawling
- ☐ staying put

MY ZONE GUESS

- ☐ sunlight
- ☐ twilight
- ☐ midnight
- ☐ abyss
- ☐ trenches
- ☐ intertidal

ANIMAL _____

REAL ZONE _____

WAS MY GUESS RIGHT? _____

HOW IT GETS FOOD _____

BEST ADAPTATION _____

What was it doing? ☐ eating ☐ hunting ☐ hiding ☐ glowing ☐ swimming ☐ resting ☐ other

Why did I guess that zone?

_____

_____

_____

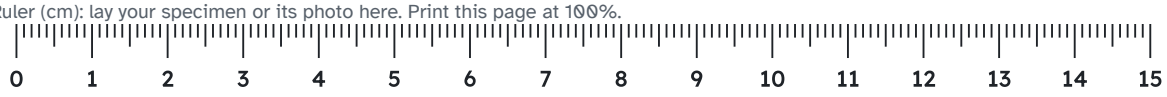
I wonder...

_____

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Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



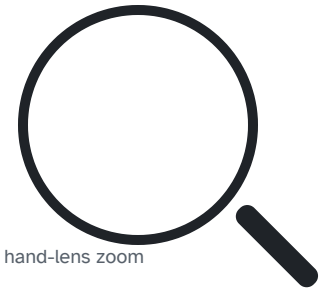
Ocean Animal #6

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

Date _____ **Place or video** _____

Where I saw it: ☐ aquarium ☐ museum ☐ beach ☐ video ☐ book ☐ other

Sketch it BIG. Label 3 adaptations.



hand-lens zoom

Real size: about _____ cm

BODY CLUES

- ☐ big eyes
- ☐ tiny or no eyes
- ☐ red or black
- ☐ see-through
- ☐ silver sides
- ☐ countershaded
- ☐ glows
- ☐ huge mouth
- ☐ soft body
- ☐ shell or armor

IT MOVES BY

- ☐ swimming
- ☐ drifting
- ☐ crawling
- ☐ staying put

MY ZONE GUESS

- ☐ sunlight
- ☐ twilight
- ☐ midnight
- ☐ abyss
- ☐ trenches
- ☐ intertidal

ANIMAL _____

REAL ZONE _____

WAS MY GUESS RIGHT? _____

HOW IT GETS FOOD _____

BEST ADAPTATION _____

What was it doing? ☐ eating ☐ hunting ☐ hiding ☐ glowing ☐ swimming ☐ resting ☐ other

Why did I guess that zone?

_____

_____

_____

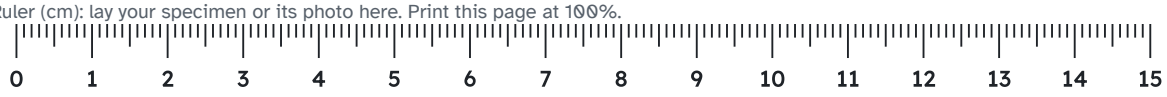
I wonder...

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_____

Ruler (cm): lay your specimen or its photo here. Print this page at 100%.



Glow Hunt: Bioluminescence

Bioluminescence (by-oh-loo-mih-NESS-ense) is light made by a living thing. Most deep-sea animals can glow! Find an animal for each way of using light, from our lessons, a book, or a video.

Way to use light	Animal	Zone
Lure prey a glowing bait draws food close		
Hide belly lights match the light above, so no shadow		
Startle predators a flash or glowing ink, then escape		
Find mates light patterns say 'I'm over here'		

Draw a glowing animal. Color the water dark and leave its lights white or yellow.

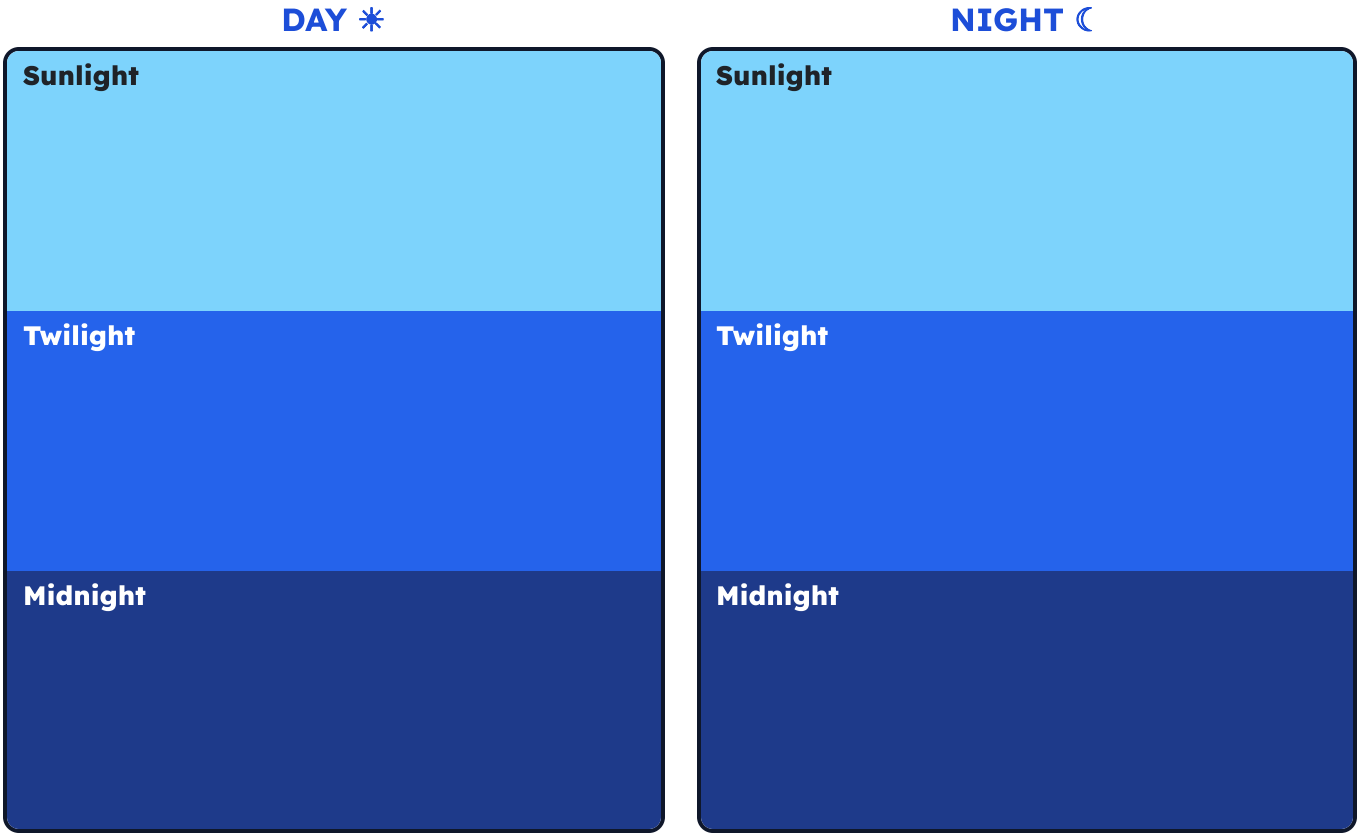
Glow near home

Fireflies are bioluminescent too! Some mushrooms, and tiny plankton in NC ocean waves, glow in the dark.

Why is glowing so useful in the deep sea?

Up at Night, Down by Day

Every night, billions of animals swim up from the twilight zone to eat near the surface, then swim back down at sunrise. It is the biggest migration on Earth: **diel vertical migration**.



Draw a lanternfish and some zooplankton where they are in the day and at night. Add arrows to show which way they swim.

Why go up at night?

Why go back down by day?

Creature Creator

Invent a brand-new deep-sea animal (use it with Independent Lab I2). Give it the adaptations it needs to survive in its zone.

Animal name _____ **Zone** _____

Draw your creature. Label every adaptation.

Its adaptations

- ☐ Finds food by: _____
- ☐ Hides from hunters by: _____
- ☐ Handles cold and pressure by: _____
- ☐ Uses light (or not) to: _____
- ☐ Finds a mate by: _____

Test a partner

Show your drawing to someone. Can they guess the zone from its clues?

☐ They guessed right! ☐ They needed a hint

A day in the life of my creature:

Beachcomber and Tide Pool Explorer

Mission: at an NC beach, sound, or aquarium touch tank, find intertidal animals and signs of them. Look at low tide along the wet sand, rocks, and jetties.

Checklist

- [ ] Ghost crab
- [ ] Barnacles
- [ ] Sea star
- [ ] Mussels or oysters
- [ ] Moon snail or whelk shell
- [ ] Sand dollar
- [ ] Jellyfish (don't touch!)
- [ ] Seaweed
- [ ] Horseshoe crab molt
- [ ] Coquina clams

Sketch my best find

Surviving the tide

Intertidal animals get pounded by waves and dried out twice a day. How does each one survive?

Animal	Its trick for waves or drying out
Barnacle	
Crab	
Sea star	
Mussel	

Museum Day: Ocean Animals

Saturday, November 7. Meet at 10 am at the Daily Planet Cafe. Look for ocean animals, living or preserved, and guess their zones.

Ocean animals I saw

Animal	Zone

A question for a museum scientist

_____

_____

_____

Sketch the coolest ocean animal

Name: _____ Adaptation: _____

Best thing I learned today

_____

_____

_____

Field Missions

One mission for each Deep Blue Sea meeting. When you finish, show a coach or your Team Captain for initials, then color in the badge.

When	Mission	Badge	Done	Initials
After Fri Oct 9 Lesson 1	Say the 5 zones in order with depths, and fill in the ocean column (p. 3).	Zone Diver	☐	
Sat Nov 7 Museum	Find 2 ocean animals and fill in the Museum Day page.	Museum Explorer	☐	
After Fri Nov 13 Lesson 2	Match the adaptations (p. 6) and log 2 Ocean Animal pages.	Adaptation Detective	☐	
After Fri Jan 8 Lesson 3	Glow Hunt: find an animal for all 4 ways to use light (p. 13).	Glow Hunter	☐	
After Fri Feb 5 Lesson 4	Draw day and night migration (p. 14). Invent a creature (Lab I2, p. 15).	Migration Tracker / Creature Creator	☐	
Fri Mar 5 Lesson 5	Cover the tags on p. 4-5 and name 10 animals' zones in under 2 minutes.	Speed Spotter	☐	
Anytime Bonus	Beachcomber page (p. 16). Log 4 Ocean Animal pages.	Beachcomber / Field Scientist	☐	

Tournament day: Sat Mar 13, Chatham Regional

- ☐ 2 sharp pencils (no notes allowed!)
- ☐ Your flashcard ring for the car ride
- ☐ A good night's sleep

- **No notes:** quiz your flashcards every other day.
- Unknown animal? Use its body clues to predict the zone.
- Remember: red looks black in the deep.

My Badges

Color in each badge when you earn it. Earn all 9 to become a **Master Ocean Explorer**! Your Team Badge Passport has the same badges.



Zone Diver

5 zones in order with depths

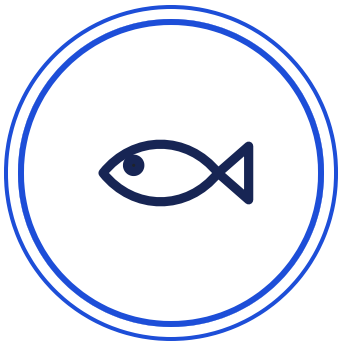
Initials: _____



Museum Explorer

Finish the Museum Day page

Initials: _____



Adaptation Detective

Match adaptations + 2 animal pages

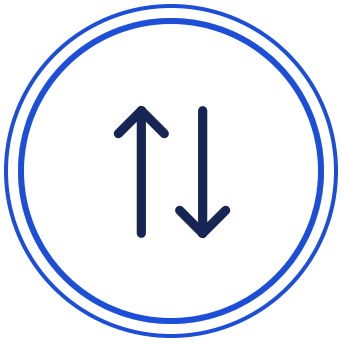
Initials: _____



Glow Hunter

4 ways animals use light

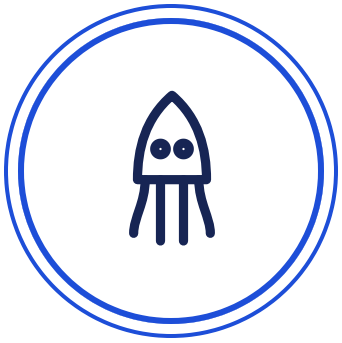
Initials: _____



Migration Tracker

Draw day and night migration

Initials: _____



Creature Creator

Invent a deep-sea animal

Initials: _____



Speed Spotter

10 zones in under 2 minutes

Initials: _____



Beachcomber

Explore the intertidal zone

Initials: _____



Field Scientist

Log 4 Ocean Animal pages

Initials: _____

My Life List

Every ocean animal you identify goes here, even from a video or a book.

#	Date	Animal	Zone	Where I saw it	Coolest adaptation
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					

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Journal made for the DDMES Science Olympiad team, 2026-27. Zones and animals match the Deep Blue Sea lesson packets and the 2027 Division A event rules.